Figures that preserve evidence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Figures that preserve evidence before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Negative results and claim revision

Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Inspecting failures and anomalies

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Power and resource tradeoffs

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Figures that preserve evidence trace and record the first step you can explain without help.

Use W2: Trace the worked case for Figures that preserve **evidence**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Figures that preserve evidence explanation against the supplied facts before discussing it.

Use the worked case for Figures that preserve **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

CountsA8/10, B9/15 are plotted as8 versus9 without denominators. Correct.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Figures that preserve evidence facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Figures that preserve evidence facts and keep your reasoning.

RatesA**80%**, B82% are plotted on bars starting79%, making B look several times larger. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Figures that preserve evidence facts and keep your reasoning.

A report omits this qualification: Visual choices must reflect the intended metric and cannot imply causal certainty from association.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Figures that preserve evidence facts and keep your reasoning.

CountsA8/10, B9/15 are plotted as8 versus9 without denominators. Correct.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Figures that preserve evidence facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Figures that preserve evidence to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled figures that preserve **evidence evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of figures that preserve **evidence**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Figures that preserve evidence

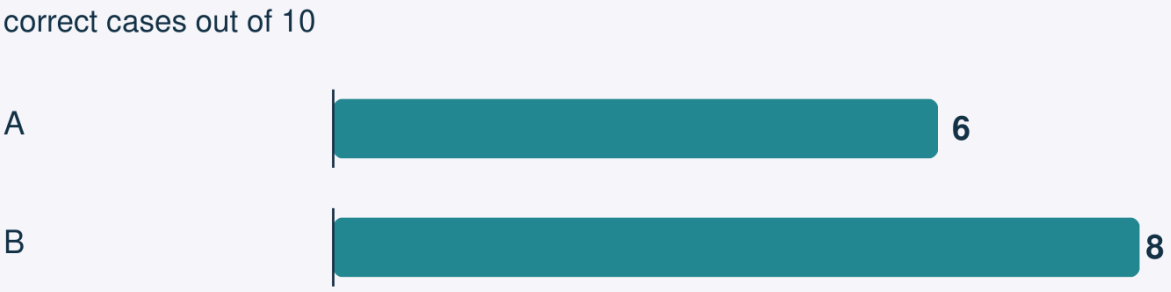
> Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases. Explain the labelled step before checking the changed condition.

Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.

A has 6/10 correct; B has 8/10. The count axis should start at zero and identify the ten-case denominator.

AI ACADEMY / AI-7/29

Zero-based correct counts



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/29

Figures that preserve evidence

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Draw honest bars

Use heights 6 and 8 on a zero-based correct-count axis.



2. Label the scale

State counts out of ten, not an unspecified quality score.



3. Keep context nearby

Note the fixed tasks and absence of a significance test.

CHANGE ONE THING

Start the axis at 5 without making the truncation obvious.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Draw count bars for 6/10 and 8/10 using the stated axis. Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Note the fixed tasks and absence of a significance test.

1. Draw honest bars

2. Label the scale

3. Keep context nearby

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Keep context nearby Note the fixed tasks and absence of a significance test. Figures make **evidence** easier to inspect when labels and scales preserve meaning.

A dramatic-looking visual cannot supply missing uncertainty or representativeness.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Start the axis at 5 without making the truncation obvious. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The bars exaggerate the visual contrast and require correction or clear disclosure.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The bars exaggerate the visual contrast and require correction or clear disclosure. Visible heights no longer correspond to the full count scale.

Repair a missing building block

Review Negative results and claim revision (ai-7/28). Goal: Explain the **mechanism** and terminology of negative results and claim revision. Short practice: Does a 6/8 tie support the **prediction** that B scores higher?

Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.. Return to this week and explain what you changed.

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the **mechanism** and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10.

What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the **mechanism** and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

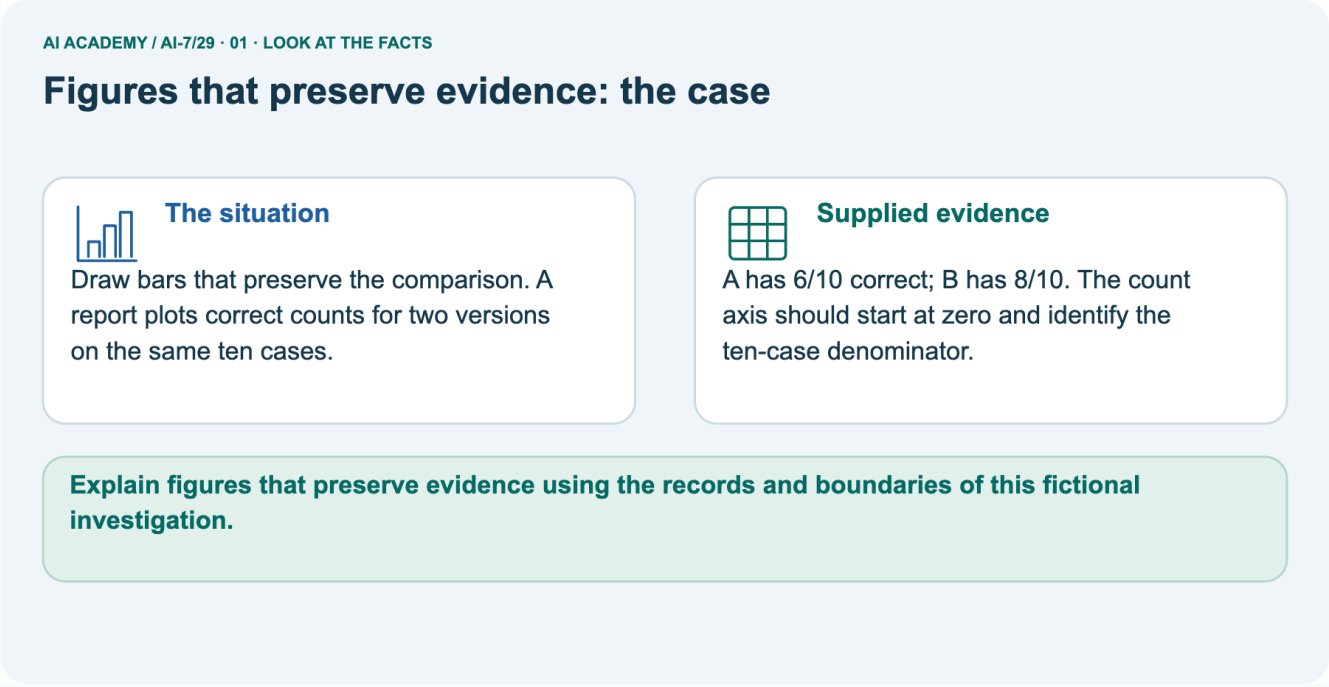
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Figures that preserve evidence picture from its labels, then record a relevant limit.

01. Figures that preserve evidence: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

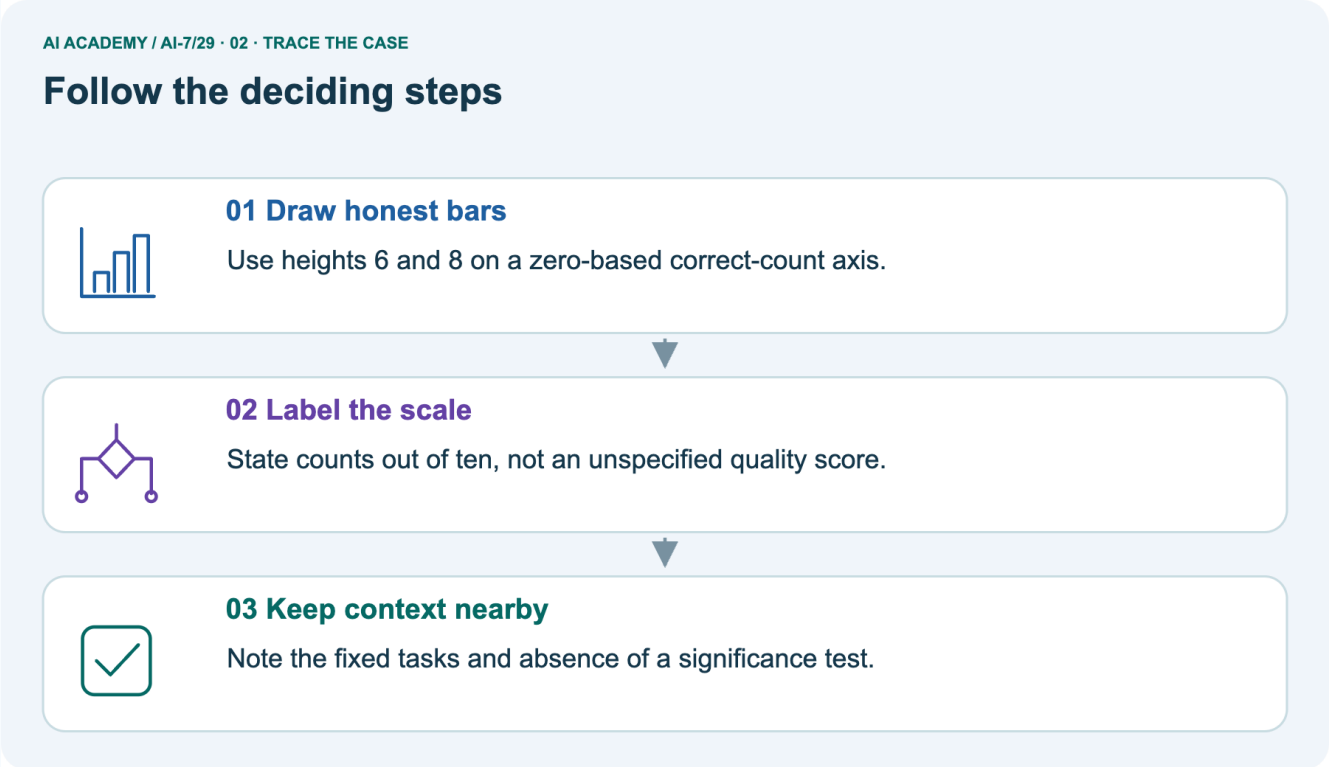
Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases. Supplied facts: A has 6/10 correct; B has 8/10.

The count axis should start at zero and identify the ten-case denominator.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



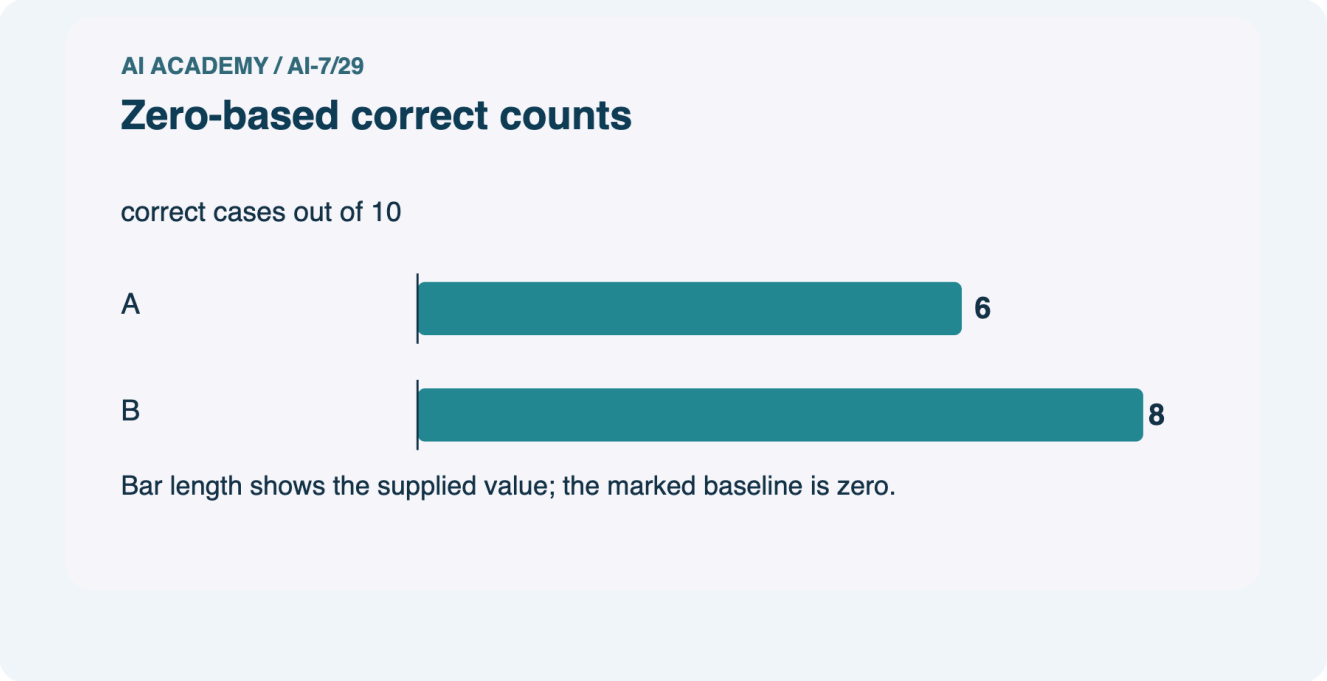
Follow the labelled picture with the matching explanation. Open the vector diagram

Draw honest bars: Use heights 6 and 8 on a zero-based correct-count axis. → Label the scale: State counts out of ten, not an unspecified quality score. → Keep context nearby: Note the fixed tasks and absence of a significance test.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Zero-based correct counts



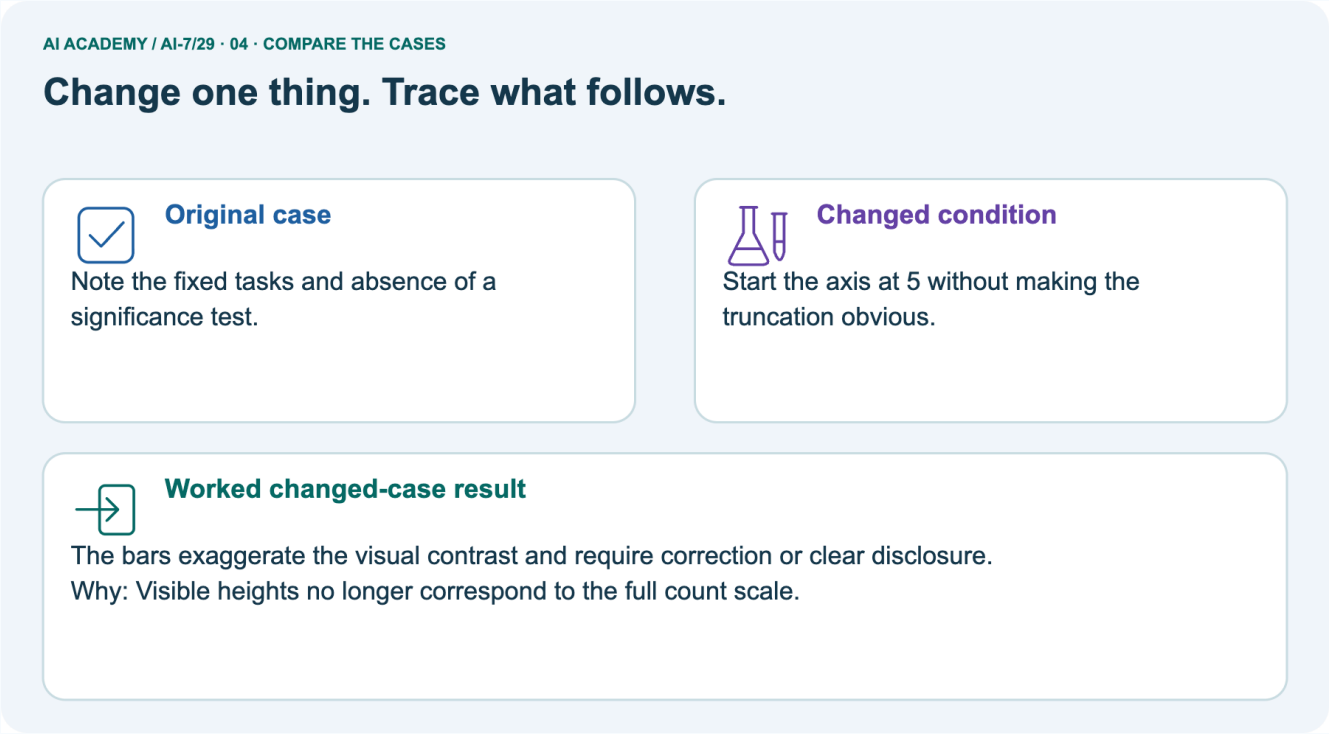
Follow the labelled picture with the matching explanation. Open the vector diagram

Zero-based correct counts. Units: correct cases out of 10. A: 6. B: 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

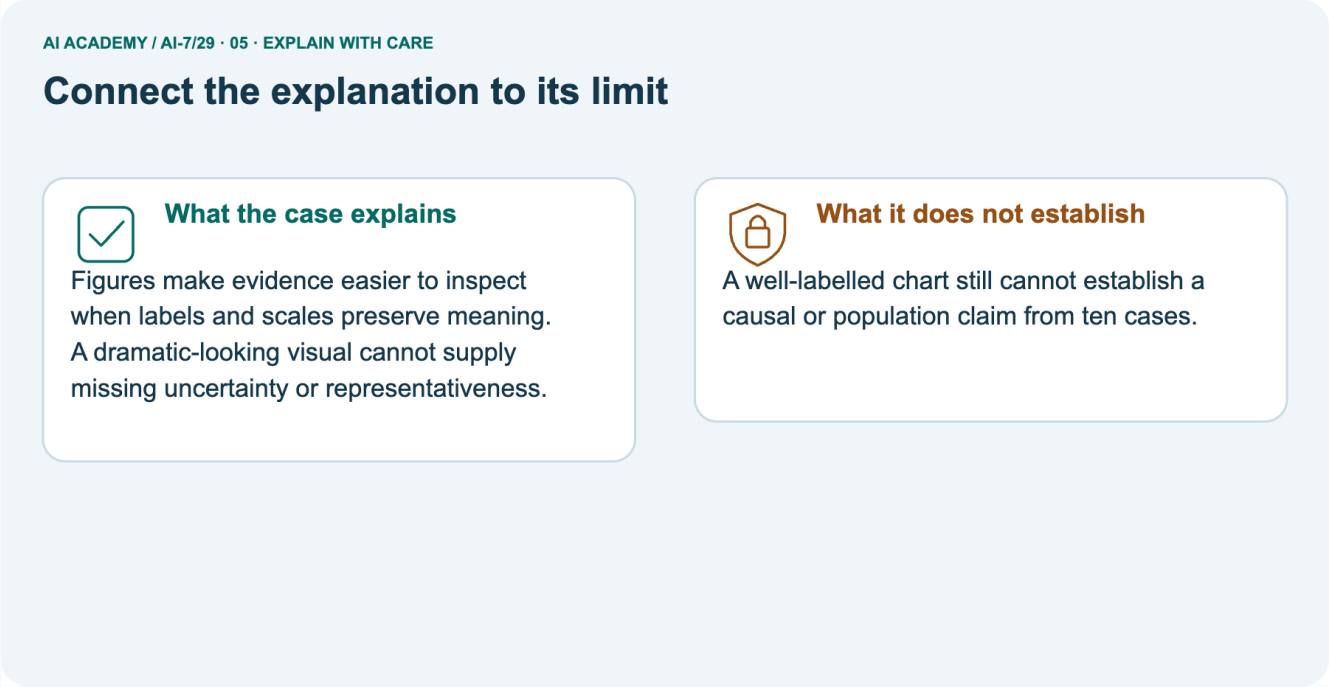
Original case: Note the fixed tasks and absence of a significance test. Changed condition: Start the axis at 5 without making the truncation obvious.

Expected result: The bars exaggerate the visual contrast and require correction or clear disclosure. Reason: Visible heights no longer correspond to the full count scale.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

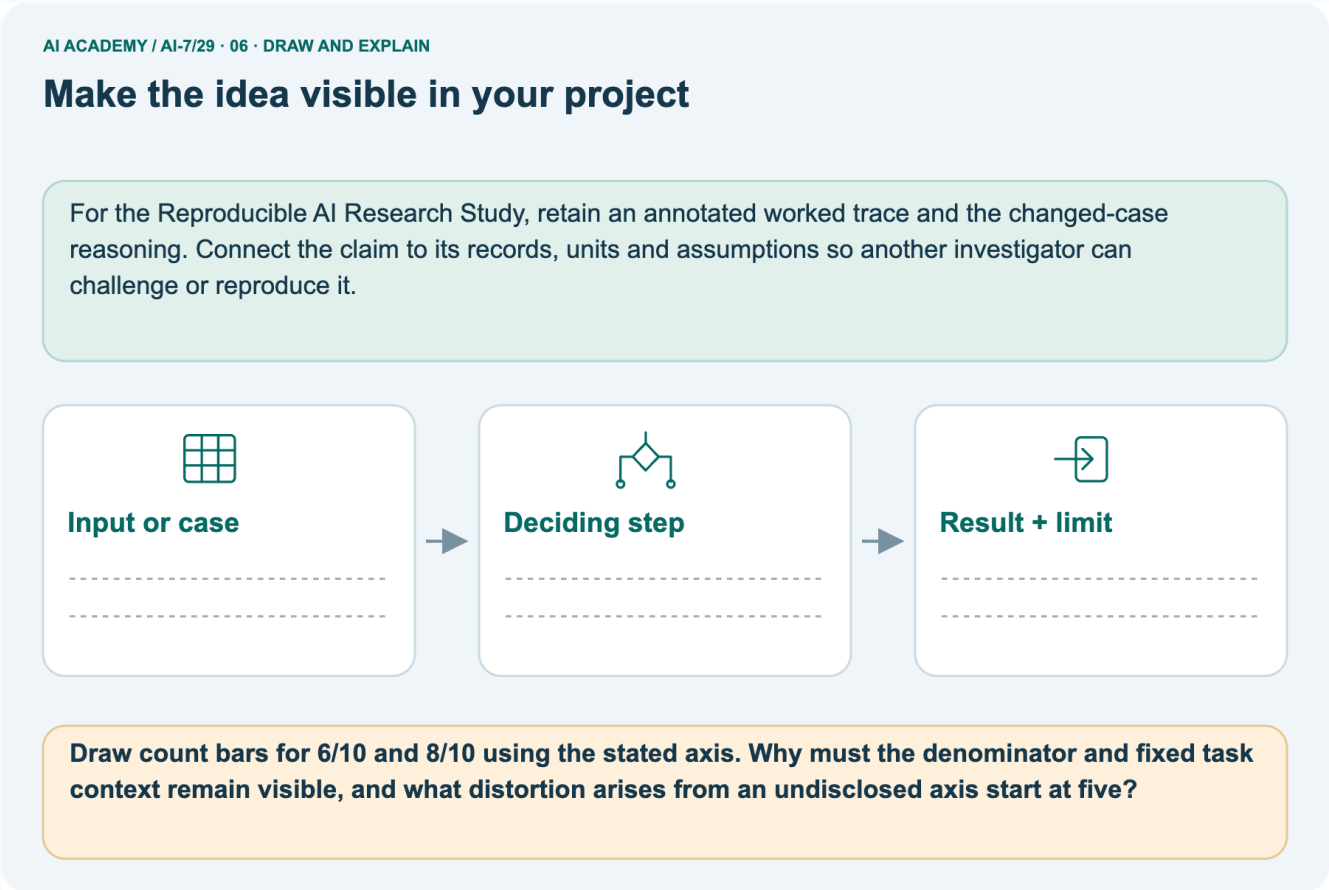
Figures make **evidence** easier to inspect when labels and scales preserve meaning. A dramatic-looking visual cannot supply missing uncertainty or representativeness.

Limit: A well-labelled chart still cannot establish a causal or population claim from ten cases.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.